

John Serences

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Education

2005: Johns Hopkins University, Baltimore, PhD, Psychological and Brain Sciences
1999: University of California, San Diego, BS, Psychology

Academic Positions

2023-present Chair, Department of Psychology, University of California, San Diego
2015-present Professor, Department of Psychology, University of California, San Diego
2011-2015 Associate Professor, Department of Psychology, University of California, San Diego
2008-2011 Assistant Professor, Department of Psychology, University of California, San Diego
2007-2008 Assistant Professor, Department of Cognitive Sciences, University of California, Irvine
2005-2006 Postdoctoral Fellow, Systems Neurobiology Lab-B, Salk Institute for Biological Studies (advisor: Geoffrey Boynton)
2000-2005: Graduate Student, Department of Psychological and Brain Sciences, Johns Hopkins University (advisor: Steven Yantis)

Awards

2018 Chancellor's Award for Excellence in Postdoctoral Scholar Mentoring
2015 Young investigator award, Vision Sciences Society
2012 James S McDonnell Foundation Scholar Award
2009 Editor's Choice award for best paper in the journal Neuroimage, Systems Neuroscience category (for: Estimating the influence of attention on population response profiles).
2004 Dean's Teaching Fellowship, Johns Hopkins University

Publications

- 1) Yantis, S., Schwarzbach, J., Serences, J.T., Carlson, R.L., Steinmetz, M.A., Pekar, J.J., Courtney, S.M. (2002). Transient neural activity in human parietal cortex during spatial attention shifts. *Nature Neuroscience*, 5(10): 995-1002
- 2) Yantis, S., Serences, J.T. (2003). Neural mechanisms of space-based and object based attentional control. *Current Opinion in Neurobiology*, 13, 187-193
- 3) Awh, E., Matsukura, M., Serences, J.T. (2003) Top-Down Control of Biased Competition During Covert Spatial Orienting. *Journal of Experimental Psychology: Human Perception and Performance*, Vol. 29(1): 52–63
- 4) Liu, T., Slotnick, S., Serences, J.T., Yantis, S. (2003). Cortical mechanisms of feature-based attentional control. *Cerebral Cortex*, 2003 Dec;13(12):1334-43
- 5) Awh, E., Serences, J., Laurey, P., Dhaliwal, H., van der Jagt, T., Dassonville, P. (2004). Evidence against a central bottleneck during the attentional blink: Multiple channels for configural and featural processing. *Cognitive Psychology*, 48(1): 95-126
- 6) Serences, J.T. (2004) A comparison of methods for estimating the event-related BOLD signal in rapid fMRI. *Neuroimage*, 21(4):1690-1700
- 7) Serences, J.T, Schwarzbach, J., Courtney, S.M., Golay, X., Yantis, S. (2004) Control of object based attention in human cortex. *Cerebral Cortex*, 14(12):1346-57
- 8) Serences, J.T., Yantis, S., Culbertson, A., Awh, E. (2004) Preparatory activity in visual cortex indexes distractor suppression during covert spatial orienting. *Journal of Neurophysiology*, 92(6):3538- 45

- 9) Serences, J.T., Shomstein, S., Leber, A.B., Golay, X., Egeth, H.E., Yantis, S. (2005) Coordination of voluntary and stimulus-driven attentional control in human cortex. *Psychological Science*, 16(2):114-122
- 10) Serences, J.T., Liu, T., Yantis, S. (2005) Parietal mechanisms of switching and maintaining attention to locations, features, and objects. *Neurobiology of Attention*, Eds. Laurent Itti, Geraint Rees, and John Tsotsos. San Diego, CA : Elsevier
- 11) Lu, H., Basso, G., Serences, J.T., Yantis, S., Golay, X., van Zijl, P.C. (2005) Retinotopic mapping in the human visual cortex using vascular space occupancy-dependent functional magnetic resonance imaging. *Neuroreport*. 16(15):1635-1640
- 12) Serences, J.T., Yantis, S. (2006) Selective visual attention and perceptual coherence. *Trends in Cognitive Science*. 10(1):38-45.
- 13) Roth, J.K., Serences, J.T., Courtney, S.M. (2006) Neural System for Controlling the Contents of Object Working Memory in Humans. *Cerebral Cortex*, Nov;16(11):1595-603.
- 14) Proulx, M., Serences, J.T. (2006) Searching for an oddball: Neural correlates of singleton detection mode in parietal cortex. *Journal of Neuroscience*, Dec 6;26(49):12631-2 (Brief Commentary Article)
- 15) Serences, J.T., Yantis, S. (2007) Spatially-selective representations of voluntary and stimulus-driven attentional priority in human occipital, parietal, and frontal cortex. *Cerebral Cortex*, Feb;17(2):284-93.
- 16) Serences, J.T., Boynton, G.M. (2007) Feature-based attentional modulations in the absence of direct visual stimulation. *Neuron* Jul 19;55(2):301-12.
- 17) Dilks, D.D., Serences, J.T., Rosenau, B., Yantis, S., McCloskey, M. (2007) Human adult cortical reorganization and consequent visual distortion. *Journal of Neuroscience* Sep 5;27(36):9585-94.
- 18) Serences, J.T., Boynton, G.M. (2007) The representation of behavioral choice for motion in human visual cortex. *Journal of Neuroscience*, Nov 21;27(47):12893-9.
- 19) Kelley, T.A., Serences, J.T., Giesbrecht, B., Yantis, S. (2008) Cortical mechanisms for shifting and holding visuospatial attention, *Cerebral Cortex*, Jan;18(1):114-25.
- 20) Mednick, S., Drummond, S., Boynton, G.M., Awh, E., Serences, J.T. (2008) Sleep-Dependent Learning and Practice-Dependent Deterioration in a Visual Crowding Paradigm. *Behavioral Neuroscience*, Apr;122(2):267-72.
- 21) Casanova, R., Ryali, S., Serences, J., Laurienti, P.J., Maldjian, J. (2008) The impact of temporal regularization on estimates of the BOLD hemodynamic response function: a comparative analysis. *Neuroimage*, May 1;40(4):1606-18.
- 22) Serences, J.T., Scolari, M., Awh, E. (2008) On-line response-selection and the attentional blink: multiple-processing channels. *Visual Cognition*; 17(4):531-554.
- 23) Serences, J.T. (2008) Value-based modulations in human visual cortex. *Neuron* Dec 26;60(6):1169-81
- 24) Serences, J.T., Saproo, S., Scolari, M., Ho, T., Muftuler, T. (2009) Estimating the influence of attention on population response profiles. *Neuroimage*, Jan 1;44(1):223-31.
- 25) Serences, J.T., Ester, E., Vogel, E., Awh, E. (2009) Stimulus-specific delay activity in human primary visual cortex. *Psychological Science*, Feb; 20(2):207-14.
- 26) Hickok, G., Okada, K., Serences, J.T. (2009) Area Spt in the human planum temporale supports sensory-motor integration for speech processing. *Journal of Neurophysiology*, May;101(5):2725-32. Epub 2009 Feb 18.
- 27) Gál, V., Kozák, K.L., Bankó, É., Kóbor, I., Serences, J., Vidnyánszky, Z. (2009) Learning to filter out visual distractors. *European Journal of Neuroscience*, Apr;29(8):1723-31.
- 28) Ho, T.C., Brown, S.D., Serences, J.T. (2009) Supramodal mechanisms of perceptual decision making in human cortex. *Journal of Neuroscience*. Jul 8;29(27):8675-87.
- 29) Scolari, M., Serences, J.T. Adaptive allocation of attentional gain. (2009) *Journal of Neuroscience*, Sep 23;29(38):11933-42.
- 30) Ester, E.E., Serences, J.T., Awh, E. (2009) Spatially global representations in human primary visual cortex during working memory maintenance. *Journal of Neuroscience*, Dec 2;29(48):15258-65.
- 31) Gál, V., Kóbor, I., Kozák, L.R., Bankó, E.M., Serences, J., Vidnyánszky, Z. (2010) Electrophysiological correlates of learning-Induced modulation of visual motion processing in humans. *Frontiers in Human Neuroscience*, Jan 6;3:69.
- 32) Okado, K., Rong, F., Venezia, J., Matchin, W., Saberi, K., Serences, J.T., Hickok, G. (2010) Hierarchical organization of human auditory cortex: evidence from acoustic invariance in the response to intelligible speech. *Cerebral Cortex*, 20(10):2486-95.

- 33) Serences, J.T. and Saproo, S. (2010). Value improves the integrity of stimulus representations in early visual cortex. *Journal of Neurophysiology*, 104(1):76-87.
- 34) Saproo, S., Serences, J.T. (2010). Influence of spatial attention on the information content of population codes in early visual cortex. *Journal of Neurophysiology*, 104(2):885-95.
- 35) Scolari, M., Serences J.T. (2010) Basing perceptual decisions on the most informative sensory neurons. *Journal of Neurophysiology*, 104(4):2266-73.
- 36) Greenberg, A., Esterman, M., Wilson, E., Serences, J.T., and Yantis, S. (2010). Control of Spatial and Feature-Based Attention in Frontoparietal Cortex. *Journal of Neuroscience*, 30(43):14330-9.
- 37) Forstmann BU, Wagenmakers EJ, Eichele T, Brown S, Serences JT. (2011) Reciprocal relations between cognitive neuroscience and formal cognitive models: opposites attract? *Trends Cogn Sci.*, 15(6):272-9. PMID: 21612972
- 38) Van Maanen, L., Brown, S.D., Eichele, T, Wagenmakers, E.J., Ho, T.C., Serences, J.T., Forstmann., B. (2011). Neural correlates of trial-to-trial fluctuations in response caution. *J Neurosci.* 31(48):17488-17495.
- 39) Serences, J.T. (2011) Mechanisms Of Selective Attention: Response Enhancement, Noise Reduction, And Efficient Pooling Of Sensory Responses. *Neuron*. 2011 Dec 8;72(5):685-7. (preview piece on a paper by Pestilli et al. in the same issue).
- 40) Serences JT., Saproo, S. (2012) Computational advances towards linking BOLD and behavior. *Neuropsychologia*. 50(4):435-46.
- 41) Scolari, M., Byers, A., Serences JT (2012) Optimal deployment of attentional gain during fine discriminations. *Journal of Neuroscience*, May 30;32(22):7723-33.
- 42) Ho, TC., Brown, SD., van Maanen L., Forstmann, BU., Wagenmakers, EJ., Serences, JT. (2012). The optimality of sensory processing during the speed-accuracy tradeoff. *Journal of Neuroscience*, Jun 6;32(23):7992-8003
- 43) Byers A, Serences JT. (2012 epub ahead of print) Exploring the relationship between perceptual learning and top-down attentional control. *Vision Research*, Jul 28
- 44) Ho TC, Brown S, Abuyo NA, Ku EH, Serences JT. (2012) Perceptual consequences of feature-based attentional enhancement and suppression. *Journal of Vision*, Aug 24;12(8)
- 45) Ester EF, Anderson DE, Serences JT, Awh E (2013) A Neural Measure of Precision in Visual Working Memory. *Journal of Cognitive Neuroscience*, May;25(5):754-61.
- 46) Garcia, JO, Srinivasan, R, Serences, JT (2013) Measuring near real-time feature-selective neural modulation in human cortex. *Current Biology*, Mar 18; 23(6):515-22. PMID: 23477721
- 47) Itthipuripat S, Garcia JO, Serences JT (2013) Temporal Dynamics of Divided Spatial Attention. *Journal of Neurophysiology*
- 48) Bridwell DA, Hecker EA, Serences JT, Srinivasan R. (2013) Individual differences in attention strategies during detection, fine discrimination, and coarse discrimination. *Journal of Neurophysiology*. Aug;110(3):784-94
- 49) Sprague, T.C., Serences, J.T. (2013) Attention modulates spatial priority maps in human occipital, parietal, and frontal cortex. *Nature Neuroscience*. Dec;16(12):1879-87.
- 50) Postle, B., Awh, E., Serences, J.T., Sutterer, D., D'Esposito, M. (2013) The positional-specificity effect reveals a passive-trace contribution to visual short-term memory. *PLoS One*. Dec 26;8(12):e83483
- 51) Itthipuripat, S., Garcia, J.O., Rungratsameetaweemana, N. Sprague, T.C., Serences, J.T. (2014) Changing the spatial scope of attention alters patterns of neural gain in human cortex. *Journal of Neuroscience*. Jan 1;34(1):112-23
- 52) Scolari M, Ester EF, Serences JT (2014) Feature- and object-based attentional modulation in the human visual system. *Oxford Handbook of Attention*. (editors: Sabine Kastner and Kia Nobre). Oxford University Press
- 53) Serences JT, Kastner S (2014) A multi-level account of selective attention. *Oxford Handbook of Attention*. (editors: Sabine Kastner and Kia Nobre). Oxford University Press
- 54) Serences, J.T., Kastner, S. (2014) Representations of space and feature-based attentional priority in human visual cortex. *Cognitive Neuroscience V*. (editors: Michael Gazzaniga and George Mangun)
- 55) Saproo, S., Serences, J.T. (2014) Attention improves transfer of motion information between V1 and MT. *Journal of Neuroscience*. Mar 5;34(10):3586-96
- 56) Ester, E.F., Ho, T.C., Brown, S.D., Serences, J.T. (2014) Variability in visual working memory ability limits the efficiency of perceptual decision making. *Journal of Vision*. Apr 2;14(4)
- 57) Byers, A., Serences, J.T. (2014). Enhanced attentional gain as a mechanism for generalized perceptual learning in human visual cortex. *Journal of Neurophysiology*. 1;112(5):1217-27.

- 58) Itthipuripat S, Ester EF, Deering S, Serences JT. (2014b) Sensory gain outperforms efficient readout mechanisms in predicting attention-related improvements in behavior. *Journal of Neuroscience*. 34(40):13384-98.
- 59) Sprague TC, Ester EF, Serences JT. (2014) Reconstructions of information in visual spatial working memory degrade with memory load. *Current Biology*. 24(18):2174-80.
- 60) Sprague, T.C., Serences, J.T. (2015) Using human neuroimaging to examine top-down modulation of visual perception. An introduction to model-based neurosciences. (editors, E.J. Wagenmakers and Birte Forstmann). Springer
- 61) Ester, E.F., Zilber, E., Serences, J.T. (2015). Substitution and pooling in visual crowding induced by similar and dissimilar distractors. *Journal of Vision*. Jan 8;15(1):15
- 62) Sprague, T.C., Saproo, S. & Serences, J.T. (2015). Visual attention mitigates information loss in small- and large-scale neural codes. *Trends in Cognitive Sciences*. Apr;19(4):215-226
- 63) Itthipuripat, S., Cha K., Rangsipat N., Serences J.T. (2015). Value-based attentional capture influences context dependent decision-making. *Journal of Neurophysiology*. 114(1):560-9.
- 64) Ester, E., Sprague, TC, Serences JT (2015) Parietal and frontal cortex encode stimulus-specific mnemonic representations during visual working memory. *Neuron*. Aug 19;87(4):893-905
- 65) Itthipuripat S, Serences JT. (2015) Integrating Levels of Analysis in Systems and Cognitive Neurosciences: Selective Attention as a Case Study. *The Neuroscientist*. 2015 Aug 25
- 66) Foster JJ, Sutterer DW, Serences JT, Vogel EK, Awh E. (2016) The topography of alpha-band activity tracks the contents of spatial working memory. *J Neurophysiol*. 1;115(1):168-77.
- 67) Sprague, T., Ester, E., & Serences, J. (2016). Restoring latent visual working memory representations in human cortex. *Neuron*. Aug 3;91(3):694-707.
- 68) Ester, E., Sutterer, D., Serences, J., & Awh, E. (2016). Feature-selective attentional modulations in human frontoparietal cortex. *The Journal of Neuroscience*. Aug 3;36(31):8188-99.
- 69) Serences JT. (2016) Neural mechanisms of information storage in visual short-term memory. *Vision Res*. 2016 Nov;128:53-67.
- 70) Bullock T, Elliott JC, Serences JT, Giesbrecht B. (2017) Acute Exercise Modulates Feature-selective Responses in Human Cortex. *J Cogn Neurosci*. Apr;29(4):605-618.
- 71) Vo V, Sprague T, Serences JT. (2017) Spatial tuning shifts increase the discriminability and fidelity of population codes in visual cortex. *The Journal of Neuroscience*. Mar 22;37(12):3386-3401
- 72) Foster JJ, Sutterer DW, Serences JT, Vogel EK & Awh E. (2017) Spatially-specific alpha-band oscillations track the latency of covert spatial orienting. *Psychological Science*. Jul;28(7):929-941
- 73) Rademaker RL, Serences JT. (2017) Pinging the brain to reveal hidden memories. *Nature Neuroscience* May 25;20(6):767-769
- 74) Itthipuripat, S., Cha, K., Byers, A., Serences, JT. (2017) Two different mechanisms support selective attention at different phases of training. *PLoS Biology*. Jun 27;15(6):e2001724
- 75) Cowell RA, Leger K, Serences JT. (2017) Feature-coding transitions to conjunction-coding with progression through human visual cortex. *J Neurophysiol*. Sep 20;jn.00503.2017.
- 76) Nelli S, Itthipuripat S, Srinivasan R, Serences, JT. (2017) Fluctuations in instantaneous frequency predict alpha amplitude during visual perception. *Nature Communications*. Dec 12; 8, 2071
- 77) Nobre AC, Serences JT. (2018) Building on a Solid Baseline: Anticipatory Biases in Attention. *Trends in Neuroscience*, Mar;41(3):120-122
- 78) Sprague TC, Itthipuripat S, Vo VA, Serences JT. (2018) Dissociable signatures of visual salience and behavioral relevance across attentional priority maps in human cortex. *J Neurophysiol*. Jun 1;119(6):2153-2165
- 79) Rungratsameetaweemana N, Itthipuripat S, Salazar A, Serences JT. (2018) Expectations do not alter early sensory processing during perceptual decision making. *J Neurosci*. Jun 13;38(24):5632-5648
- 80) Itthipuripat S, Cha K, Deering S, Salazar AM, Serences JT. (2018) Having more choices changes how human observers weight stable sensory evidence. *Journal of Neuroscience*. Oct 3;38(40):8635-8649
- 81) Chunharas C, Rademaker RL, Sprague TC, Brady TF, Serences JT (2019). Separating memoranda in depth increases visual working memory precision and capacity. *Journal of Vision*. Jan 2;19(1):4.
- 82) Henderson M, Serences JT. (2019) Human frontoparietal cortex represents behaviorally relevant target status based on abstract object features. *J Neurophysiol*. Apr 1;121(4):1410-1427
- 83) Itthipuripat S, Deering S, Serences JT. (2019) When Conflict Cannot be Avoided: Relative Contributions of Early

- Selection and Frontal Executive Control in Mitigating Stroop Conflict. *Cereb Cortex*. Mar 16. pii: bhz042.
- 84) Rungratsameetaweemana N, Serences JT. (2019) Dissociating the impact of attention and expectation on early sensory processing. *Curr Opin Psychol*. Mar 23;29:181-186.
 - 85) Sutterer DW, Foster JJ, Serences JT, Vogel EK, Awh E. (2019) Alpha-band oscillations track the retrieval of precise spatial representations from long-term memory. *J Neurophysiol*. Jun 12. doi: 10.1152/jn.00268.2019
 - 86) Rademaker RL, Chunharas C, Serences, JT (2019). Coexisting representations of sensory and mnemonic information in human visual cortex. *Nature Neuroscience*. Jul 1. doi: 10.1038/s41593-019-0428-x
 - 87) Itthipuripat, S., Sprague, T.C., Serences, J.T. (2019). Functional MRI and EEG index complementary attentional modulations. *Journal of Neuroscience*. May 24. pii: 2519-18. doi: 10.1523/JNEUROSCI.2519-18.2019
 - 88) Itthipuripat S, Vo V, Sprague TC, Serences JT (2019). Value-driven attentional capture enhances distractor representations in early visual cortex. *PLoS Biology*. 17(8):e3000186. doi: 10.1371/journal.pbio.3000186.
 - 89) Henderson MM, Vo V, Chunharas C, Sprague TC, Serences JT (2019). Multivariate analysis of BOLD activation patterns recovers graded depth representations in human visual and parietal cortex. *eNeuro* 0362-18.2019. doi: 10.1523/ENEURO.0362-18.2019.
 - 90) Rungratsameetaweemana N., Squire, LR, Serences, JT (2019) Preserved capacity for learning statistical regularities and directing selective attention after hippocampal lesions. *Proceedings of the National Academy of Sciences*. 116(39):19705-19710
 - 91) Adam KCS, Serences JT. (2019) Working Memory: Flexible but Finite (commentary on: *A Flexible Model of Working Memory*, by Bouchacourt and Buschman). *Neuron*. 103(2):184-185. doi: 10.1016/j.neuron.2019.06.025.
 - 92) Sprague TC, Boynton GM, Serences JT. (2019) The importance of considering model choices when interpreting results in computational neuroimaging. *eNeuro*. pii: ENEURO.0196-19.2019.
 - 93) Itthipuripat S, Chang KY, Bong A, Serences JT (2019) Stimulus visibility and uncertainty mediate the influence of attention on response bias and visual contrast appearance. *Journal of Vision*. 19(14):8.
 - 94) Ester EF, Sprague TC, Serences JT (2020) Categorical Biases in Human Occipitoparietal Cortex. *The Journal of Neuroscience*. 40(4):917-931.
 - 95) Adam KCS, Chang L, Rangan N, Serences JT (2021) Steady-State Visually Evoked Potentials and Feature-based Attention: Preregistered Null Results and a Focused Review of Methodological Considerations. *Journal of Cognitive Neuroscience*. 33(4):695-724.
 - 96) Nelli, S., Malpani, A., Boonjindasup, M., & Serences, J.T. (2021) Individual Alpha Frequency Determines the Impact of Bottom-Up Drive on Visual Processing. *Cerebral Cortex Communications*, 2(2).
 - 97) Adam, K.C.S, Patel, T., Rangan, N., and Serences, J.T. (2021) Classic visual search effects in an additional singleton task: An open dataset. *Journal of Cognition*. Jul 28;4(1):34.
 - 98) Henderson, M. and Serences, J.T. (2021). Biased orientation representations can be explained by experience with non- uniform training set statistics. *Journal of Vision*. Aug 2;21(8):10.
 - 99) Adam KCS, Serences JT. (2021) History Modulates Early Sensory Processing of Salient Distractors. *The Journal of Neuroscience*. Sep 22;41(38):8007-8022
 - 100) Vo, V.A., Sutterer, D.W., Foster, J.J., Sprague, T.C., Awh, E. and Serences, J.T. (2022). Shared representational formats for information maintained in working memory and information retrieved from long-term memory. *Cerebral Cortex*. Feb 19;32(5):1077-1092.
 - 101) Park S, Serences JT. (2022) Relative precision of top-down attentional modulations is lower in early visual cortex compared to mid- and high-level visual areas. *Journal of Neurophysiology*. Feb 1;127(2):504-518.
 - 102) Chunharas C, Rademaker RL, Brady TF, Serences JT. (2022) An adaptive perspective on visual working memory distortions. *J Exp Psychol Gen*. Feb 21.
 - 103) Henderson, M., Rademaker, R.L., Serences J.T. (2022) Flexible utilization of spatial- and motor-based codes for the storage of visuo-spatial information. *eLife*. 11:e75688
 - 104) Adam, K.C.S., Rademaker, R.L., and Serences, J.T. (2022) Evidence for, and challenges to, sensory recruitment models of visual working memory. In T.F. Brady & W.A. Bainbridge (Eds.), *Visual Memory* (pp. 5-25). Routledge.
 - 105) Sheehan TC, Serences JT. (2022) Attractive serial dependence overcomes repulsive neuronal adaptation. *PLoS Biology*. Sep 6;20(9):e3001711.106)
 - 106) Adam KCS, Rademaker RL, Serences JT. (2022) Dynamics Are the Only Constant in Working Memory. *The Journal of Cognitive Neuroscience*. Dec 1;35(1):24-26.
 - 107) Sawetsuttipan P, Phunchongharn P, Ounjai K, Salazar A, Pongsuwan S, Intrachoto S, Serences JT, Itthipuripat S.

- (2023) Perceptual Difficulty Regulates Attentional Gain Modulations in Human Visual Cortex. *The Journal of Neuroscience*. May 3;43(18):3312-3330.
- 108) Itthipuripat S, Phangwiwat T, Wiwatphonthana P, Sawetsuttipan P, Chang KY, Störmer V, Woodman G, Serences JT (2023) Dissociable neural mechanisms underlie the effects of attention on visual appearance and response bias. *The Journal of Neuroscience*. Sep 27;43(39):6628-6652.
- 109) Wennberg JW, Serences JT. (2024) Mixing and mingling in visual working memory: Inter-item competition is feature-specific during encoding and feature-general during maintenance. *Atten Percept Psychophys*. Aug;86(6):1846-1860.
- 110) Adam, K.C.S., Yang, Z., and Serences, J.T. (2024). First encounters: Estimating the initial magnitude of attentional capture. *Visual Cognition*. Nov 15;32:822-844.
- 111) Henderson MM, Serences JT, Rungratsameetaweemana N. (2025) Dynamic categorization rules alter representations in human visual cortex. *Nat Commun*. Apr 11;16(1):3459.
- 112) Park S, Serences JT. Near-random connections support top-down feature-based attentional modulations in early sensory cortex (2025). *PLoS Comput Biol*. Aug 12;21(8):e1013396
- 113) Kiyonaga A, Serences JT. (2025) Sensory reformatting for a working visual memory. *Trends Cogn Sci*. Dec;29(12):1120-1135.
- 114) Shafer-Skelton A, Brady TF, Serences JT. (2026) A 2D Gabor-wavelet baseline model out-performs a 3D surface model in scene-responsive cortex. *PLoS Comput Biol*. 2026 Feb 2;22(2):e1013888. doi: 10.1371/journal.pcbi.1013888.
- 115) Wennberg JW, Serences JT. (2026) Similarity-driven compression during encoding supports biased but more precise working memory. *J Vis*. 2026 Apr 1;26(4):8. doi: 10.1167/jov.26.4.8.
- 116) Kular H, Serences JT. (in press) Memory recall errors reflect interacting sensory and mnemonic representations. *Psychonomic Bulletin & Review*
- 117) Chaisilprungraung T, Sawetsuttipan P, Serences JT, Itthipuripat S. (in press) Training under attentional competition produces learned biases in visual appearance. *Psychonomic Bulletin & Review*

Edited Book

(2018) Sensation, Perception and Attention. Serences JT, Editor Volume II of the 5th edition of The Stevens' Handbook of Experimental Psychology and Cognitive Neuroscience. Editor in Chief Wixted JT.

Active Grants

- 1) NEI R01EY025872 (2010-present, last renewed May 2025). Title: Adaptive population codes for flexible visually-guided behaviors. Role: PI
- 2) NEI R01EY036843 (2025-present). Title: Pupillary working memory for adaptive perception. Role: CO-I (PI Anastasia Kiyonaga)

Finished grants

- 1) NIMH-2R01MH087214 (2019-2024). Title: Neural indices of online and offline states in human working memory. Role: CO-I (PIs Edward Awh and Edward Vogel, University of Chicago). Direct Costs to UCSD (annually): approx. \$130,000
- 2) NIMH RF-1MH114277-01 (2017-2021). Title: Using fMRI to Measure the Neural-level Signals Underlying Population-level Responses. Role: CO-I (PIs Rosemary Cowell and David Huber, U Mass Amherst, other CO-I Earl Miller at MIT). Effort: 2.0 months
- 3) NIMH R01-MH087214 (2014-2019). Title: Oscillatory Measures of Number and Precision in Working Memory. Role: CO-I (PIs Edward Awh and Edward Vogel at U. Chicago). Total Costs to UCSD: \$300,415
- 4) James S McDonnell Foundation Scholar Award (2012-2019). Title: Linking attentional modulations in human visual cortex with perception and behavior. Role: PI. Total Direct Costs: \$600,000
- 5) NEI R21-EY024733 (2014-2016) Title: Oscillatory dynamics and sensory processing. Role: PI. Total Direct Costs: \$275,000
- 6) NIMH R01-02345 (2010-2015) Title: Adaptive allocation of attentional gain. Role: PI. Total direct costs: \$1,250,000
- 7) Australian Research Council (2012-2014) Title: Rapid Decisions: From Neuroscience to Complex Cognitions.

Role: CO-I. PI of grant: Scott Brown, University of Newcastle, Australia

- 8) NIMH R01 (2009-2013) Title: On the distinction between number and resolution in visual working memory. Role: CO-I. PI's of grant: Edward Awh and Edward Vogel, U. Oregon. Total direct costs to UCSD: \$143,108
- 9) NIDCD R01 DC009659 (2008-2013) Title: Integrative functions of the planum temporal. Role: CO-I. PI of: Greg Hickok, UC Irvine. Total direct costs to UCSD: \$74,790
- 10) NIMH R01 68004-02 (2009-2013) Title: Dynamic Neuroimaging with High-Resolution SSVEPs. Role: CO-I. PI of grant: Ramesh Srinivasan, UC Irvine. Total direct costs to UCSD: \$218,084
- 11) Dutch organization for scientific research. Title: The Neural Basis of Decision-Making with Multiple Choice Alternatives. Role: CO-I. PIs of grant: Eric-Jan Wagenmakers, Birte Forstmann, University of Amsterdam. Total costs: 228,000 Euros (no direct costs to UCSD)
- 12) Academic Senate Research Award (2009-2010), UCSD, Neural mechanisms of perceptual learning and expertise. Role: PI. Total direct costs: \$14,700
- 13) R21-MH083902 (2008-2010). Title: Assessing space and feature based attention with fMRI and multivoxel pattern analysis. Role: PI. Total direct costs: \$275,000
- 14) NSF-0818418 (2008-2011). Title: Attention and perceptual coherence. Role: PI. Award offered by Cognitive Neuroscience panel but declined due to partial overlap with R21-MH083902

Post-doctoral/Graduate school grants

- 1) 7/06-11/06 F32EY017261, Individual National Research Service Award, National Eye Institute
- 2) 9/05-7/06 T32-MH20002. Institutional National Research Service Award, Institute for Neural Computation Training Grant, University of California, San Diego
- 3) Fall 2004 Dean's Teaching Fellowship (by competition), Johns Hopkins University, Psychology and Social Policy
- 4) 2002-2005 National Science Foundation Graduate Research Fellowship
- 5) 2001-2002 T32-EY07143, Visual Neuroscience Training Grant, Wilmer Ophthalmological Institute, Johns Hopkins University School of Medicine

Professional Affiliations

Society for Neuroscience

Vision Sciences Society

American Psychological Association

Editorial positions

2026-present Associate Editor: Vision Research

2009-present Associate Editor: Cognitive Neuroscience

2014-present Editorial Board: Journal of Neurophysiology

2019-2023 Reviewing Editor: eLife

2014-2017 Associate Editor: Psychonomic Bulletin and Review

General Academic Service

- 1) Organizer of *Computational Neuroscience: Vision* two-week summer school at Cold Spring Harbor Labs in NY (with Prof. Jonathan Pillow from Princeton and Prof. Emily Cooper from UC Berkeley) 2022-present
- 2) Chair, Neuroscience of Basic Visual Processes Study Section, National Institute of Health, October 2019 – June 2021
- 3) Standing member, Mechanisms of Sensory, Perceptual, and Cognitive Processes Study Section, National Institute of Health, October 2016 – October 2019
- 4) Scientific committee, 13th International Conference on Cognitive Neuroscience, 2016-2017
- 5) Submission reviewer, Cognitive Computational Neuroscience, 2017
- 6) Ad-hoc member, Mechanisms of Sensory, Perceptual, and Cognitive Processes Study Section, National Institute of Health, Feb 2012, Oct 2012, Oct 2013, Oct 2014, June 2015
- 7) Special ad-hoc panel review panel, National Institute of Health, July 2011 and July 2012
- 8) Sensory and Motor Neuroscience, Cognition and Perception Fellowship Study Section, National Institute of Health, March 2013, June 2013

- 9) COSYNE: Abstract reviewer 2011- 2013
- 10) Ad-hoc reviews for National Science Foundation, Medical Research Council (UK), BBSRC (UK)
- 11) Organizer, Kavli Institute for Brain and Mind Symposium, Bridging Levels Of Analysis In Neuroscience: From Single Cells To Large Scale Networks ([web link](#))
- 12) Co-organizer of NeuroCog Collective meetings (lead organizer Cory Miller), 2012, 2014, 2016
- 13) Co-organizer Southern California Conference on Learning and Memory, 2011

Department Service

- 1) Graduate Advisors Panel, 2019-present
- 2) Academic Personnel Committee, 2019-2022
- 3) Chair, Climate and Diversity Committee, 2017-2022
- 4) Space / Resources Committee, Department of Psychology, 2016-2019
- 5) Chair, faculty search committee, Computation, 2017-2018
- 6) Undergraduate affairs, Department of Psychology, 2011-2018
- 7) Colloquium/Distinguished Lecturer Series, Department of Psychology, 2016-2018
- 8) Chair, faculty search committee, Sensation and Perception, 2015-2016
- 9) Faculty search committee, Cognitive Psychology, 2014-2015
- 10) Chair, Undergraduate curriculum revision committee, Department of Psychology, 2014-2015
- 11) Diversity committee, Department of Psychology, 2013-2015
- 12) Faculty search committee, Cognitive and Behavioral Neuroscience, UCSD, 2013-2014
- 13) Faculty search committee, Social Psychology, UCSD, 2011-2012

College/University/Academic Senate Service

- 1) Chair of Graduate Admissions, Neuroscience Graduate Program 2021-2023
- 2) Representative Assembly 2021-23
- 3) Senate Council (at large member from Rep Assembly) 2021-22
- 4) PATHS reviewer, April 2021
- 5) Executive committee, Neuroscience Graduate Program 2013- present
- 6) Graduate Admissions Committee, Neuroscience Graduate Program 2016-present
- 7) Minor prop committee, Neuroscience Graduate Program, 2012-present
- 8) Chair, System-wide University Committee on Educational Policy, 2019-2020
- 9) Chair, System-wide task force on online education, 2019-2020
- 10) System-wide Academic Council, 2019-2020
- 11) System-wide Academic Planning Council, 2019-2020
- 12) System-wide task force on teaching evaluation, 2019-2020
- 13) Intersegmental Committee of Academic Senates (ICAS), 2019-2020
- 14) UCSD Academic Council, 2019-2020
- 15) UCSD Senate Administrative Council 2019-2020
- 16) Executive committee, Keck Center for Functional Magnetic Resonance Imaging 2014-present
- 17) Vice Chair, System-wide University Committee on Educational Policy, 2018-2019
- 18) Member, System-wide University Committee on Educational Policy, 2017-2018
- 19) Educational Finance Modelling, UCOP 2018-2019
- 20) Senate Ad Hoc Committee for proposed discontinuations of UG and grad degree programs, 2018
- 21) UC Committee on Educational Policy, system-wide committee, 2017-2018
- 22) Undergraduate Council, UC San Diego, 2017-2018
- 23) Creation of Cognitive and Behavioral Neuroscience major (joint with Department of Cognitive Science)
- 24) Council on Undergraduate Education 2013-2018
- 25) Undergraduate Curriculum Review Committee, Revelle College, UCSD 2015
- 26) Math curriculum revision sub-committee, Revelle College, UCSD 2014-2015
- 27) Curriculum revision committee for CSE3 (computing literacy course) at UCSD, 2008-2009

Ad-hoc Journal Reviewing (partial list)

Cerebral Cortex, Cognitive Brain Research, Cognition, Current Biology, European Journal of Neuroscience, Experimental Brain Research, eLife, Human Brain Mapping, Journal of Cognitive Neuroscience, Journal of Experimental Psychology: Human Perception and Performance Journal of Experimental Psychology: General
Journal of Neurophysiology Journal of Neuroscience Journal of Vision, Nature, Nature Neuroscience, Neuron, Neuroimage, Neuropsychopharmacology, Neuroreport, Perception, Perception and Psychophysics, Psychonomic Bulletin & Review, Psychological Review, Psychological Science, PLoS Computational Biology, Proceedings of the National Academy of Sciences, Progress in Neuro-Psychopharmacology & Biological Psychiatry, Trends in Cognitive Science, Trends in Neuroscience, Vision Research

Invited Lectures

3/2026 Ernst Strüngmann Institute der Max-Planck-Gesellschaft
12/2025 University of California, Santa Barbara
11/2024 Johns Hopkins University
4/2021 Cognitive Brown Bag, Dartmouth
9/2020 Colloquium, Texas A&M
9/2020 Lecture, "Vision in Context" workshop, Donders Institute, Netherlands
3/2019 Colloquium, Georgia Tech University
1/2019 Colloquium, George Washington University
7/2018 Visual Search and Selective Attention, Munich, Germany
4/2018 Cognitive Neuroscience Colloquium, UC Berkeley
3/2018 CPCN/N&B Colloquium, UC Santa Barbara
1/2018 Colloquium, Brown University Department of Cognitive, Linguistic and Psychological Sciences
5/2017 Colloquium, Core for Advanced MRI (CAMRI) at Baylor College of Medicine
5/2017 Psychology Department Colloquium, Boston College
4/2017 Speaker at the "What is Attention?" conference, Carnegie Mellon/University of Pittsburgh
3/2017 Guest lecturer in REV 20 (Honors seminar for Revelle College, UCSD)
3/2017 Neuroscience Seminar Series, University of Texas, Austin
11/2016 Center for Cognitive Neuroscience Colloquium, Duke University
6/2016 NeuroCog IV, Nosara, Costa Rica
6/2016 Center for Visual Science, University of Rochester (Future of Attention conference)
5/2016 Pint of Science San Diego, Memory from Molecules to Minds (with co-panelist Brenda Bloodgood, Gentry Patrick, Stefan Leutgeb)
3/2016 Center for Neural Basis of Cognition, Carnegie Mellon / University of Pittsburgh
3/2016 Medical University of South Carolina Neuroscience Colloquium
1/2016 IBRO, Budapest, Hungary
10/2015 SPARKS conference, Bernstein Center for Computational Neuroscience, Berlin, Germany
4/2015 Stanford Center for Mind, Brain and Computation
3/2015 University of Nevada Reno
3/2015 Tutorial lecture on multivariate data analysis, University of Nevada Reno
12/2014 Peking University, Beijing China
10/2014 Yantis Vision Seminar, Johns Hopkins University, Baltimore, MD
8/2014 Bernstein Center for Computational Neuroscience, Berlin, Germany
7/2014 NeuroCog III, Coffs Harbor, NSW, Australia
7/2014 Computational Neuroscience: Vision, Cold Spring Harbor Labs, NY
2/2014 Cognitive Brownbag series, UCSD
11/2013 Workshop on Interfacing Models with Brain Signals, UC Irvine
11/2013 Attention and Learning workshop, San Diego (NSF funded meeting)
11/2013 University of Arizona
10/2013 Rovereto Attention Workshop, Università degli Studi di Trento
6/2013 James S McDonnell Foundation workshop in Cambridge, UK
4/2013 Symposium Speaker, Cognitive Neuroscience Society Annual Meeting, San Francisco (with Jack Gallant and Frank Tong)

3/2013 Department of Psychological and Brain Sciences, Dartmouth University
 12/2012 3rd annual Australasian Cognitive Neuroscience Society, University of Queensland, Brisbane Australia
 12/2012 University of Queensland Centre for Perception and Cognitive Neuroscience (CPCN) workshop on fMRI methods
 11/2012 Macquarie University, Sydney Australia
 10/2012 CTIR - CFMRI Joint Symposium on Functional Neuroimaging
 7/2012 Summer School in Cognitive Neuroscience, U. Leipzig
 5/2012 'Combining Cognitive Models and fMRI', Royal Netherlands Academy of Arts and Sciences
 5/2012 'New Insights from Model-based Cognitive Sciences' conference, Royal Netherlands Academy of Arts and Sciences
 3/2012 University of Oregon
 2/2012 Cognitive Brown Bag, UC Riverside
 1/2012 NeuroCog II, Nosara, Costa Rica
 10/2011 International workshop on visual attention, Allahabad, India
 12/2010 Vanderbilt University
 11/2010 Merk Neurosciences Seminar Series, UCSD
 11/2010 Symposium at Society for Neuroscience Annual Meeting (Model-based neuroimaging and decision neurosciences)
 9/2010 Panelist in the Presidential Symposium at Society for Psychophysiological Research Annual Meeting (Competing for attention: cortical mechanisms of selection)
 4/2010 Symposium speaker at Cognitive Neuroscience Society Annual Meeting (Prefrontal cortex and perceptual decision making)
 2/11/2010 Department of Psychology Colloquium, Stanford University
 2/10/2010 Department of Psychology Colloquium, Stanford University
 1/10/2010 NeuroCog, Bocas Del Torro, Panama
 11/2009 Neuroscience Institute Colloquium, Princeton University
 3/2009 International Organization of Schizophrenia Research
 11/2008 University of California, San Diego, Center for Functional Magnetic Resonance Imaging
 5/2008 Invited speaker at symposium on attention and object recognition, annual meeting of Vision Sciences Society, Naples Florida
 4/2008 University of California, Irvine, Center for Neurobiology and Behavior (CNLM) annual meeting
 2/2008 Department of Psychology Colloquium, University of California, San Diego
 2/2008 Department of Psychology Colloquium, University of California, Berkeley
 1/2008 University of California, Irvine, Social Science Dinner Club (public lecture series)
 10/2007 Cognitive Forum Colloquium, University of California, Los Angeles
 06/2007 Cognition and Perception Group Colloquium, University of California, Santa Barbara
 11/2006 Department of Psychology Colloquium, University of Oregon
 4/2006 Colloquium, Smith-Kettlewell Eye Research Institute
 2/2006 Department of Psychology Colloquium, University of Michigan
 2/2006 Department of Cognitive Science Colloquium, University of California, Irvine
 1/2006 Department of Psychology Colloquium, University of Washington
 6/2005 Department of Cognitive Science Colloquium, University of California, Irvine
 7/2004 Systems Neurobiology Lab Colloquium, Salk Institute for Biological Studies

Teaching

- 1) Computational Social Science (CSS1). First course in the core sequence for the computational social sciences minor. Focus on learning Python in the Jupyter environment and basic data wrangling/analysis skills.
- 2) PSYC 102 Sensory Neuroscience. Undergraduate core upper division course focused on the functional neuroanatomy of the five different sensory systems.
- 3) NEU231/PSYC231 Tools for experimental data analysis. Graduate course introducing Python, Jupyter Notebooks, and Git along with data analysis methods that are commonly used in neuroscience/psychology including mutual information, Fourier transforms, time/frequency domain filtering, multivariate pattern classification, granger causality and model fitting.

- 4) Cognitive and Behavioral Neuroscience Brown Bag. Weekly lecture series that provides a venue for graduate student talks from labs with a focus on neuroscience

Teaching (past)

- 1) Laboratory in the Analysis of Psychological Data (Upper division undergraduate research methods/statistics course, taught in Spring 2003 and Spring 2004 at Johns Hopkins University)
- 2) Psychology and Social Policy (Upper division undergraduate, designed course and taught in Fall 2004 at Johns Hopkins University, awarded Dean's Teaching Fellowship)
- 3) Attention and Perception (Graduate core curriculum course, Fall 2007 at UC Irvine)
- 4) Special Topics and Research in Cognitive Neuroscience (Graduate seminar, Fall 2007, Winter 2008, and Spring 2008 at UC Irvine, with Emily Grossman)
- 5) Introduction to Psychology (Undergraduate core course, Spring 2008 at UC Irvine)

Former post-doctoral associates

Kirsten Adam (Assistant Professor of Psychology, UC Davis)
 Rosanne Rademaker (Group leader, Ernst Strüngmann Institute and MPI in Frankfurt)
 Edward Ester (Assistant Professor of Psychology, University of Nevada, Reno)
 Rosemary Cowell (Associate Professor of Psychology, University of Colorado, Boulder)
 Javier Omar Garcia (US CCDC Army Research Laboratory)

Former psychology graduate students

Janna Wennberg (postdoc at UCSD in Cognitive Science)
 Sunyoung Park (postdoc at Rutgers)
 Anna Shafer-Skelton (postdoc at UT Austin)
 Angus Chapman (postdoc at Boston University)
 Chaipat Chunharas (Department of Medicine and PI of the Cognitive Clinical & Computational Neuroscience Lab, King Chulalongkorn Memorial Hospital, Chulalongkorn University, Bangkok, Thailand)
 Miranda Scolari (Associate Professor of Psychology, Texas Tech)
 Tiffany Ho (Associate Professor of Psychology, UCLA)
 Sameer Saproo (Qualcomm, Product Manager)
 Anna Byers (ServiceNow, Senior Manager UX)
 Mary Smith (Researcher at Adobe)

Former neuroscience graduate students

Margaret Henderson (Assistant Professor of Psychology, CMU).
 Timothy Sheehan (data scientist at Cumulus Neuroscience)
 Nuttida Rungratsameetaweemana (Assistant Professor of Biomedical Engineering, Columbia University)
 Stephanie Nelli (Assistant Professor of Psychology, Occidental College)
 Vy Vo (AI/ML Research Scientist at Intel Labs)
 Thomas Sprague (Associate Professor of Psychological and Brain Sciences, UC Santa Barbara)
 Sirawaj Itthipuripat (Associate Professor and Director of Neuroscience Center for Research and Innovation at Learning Institute, King Mongkut's University of Technology Thonburi, Bangkok, Thailand)